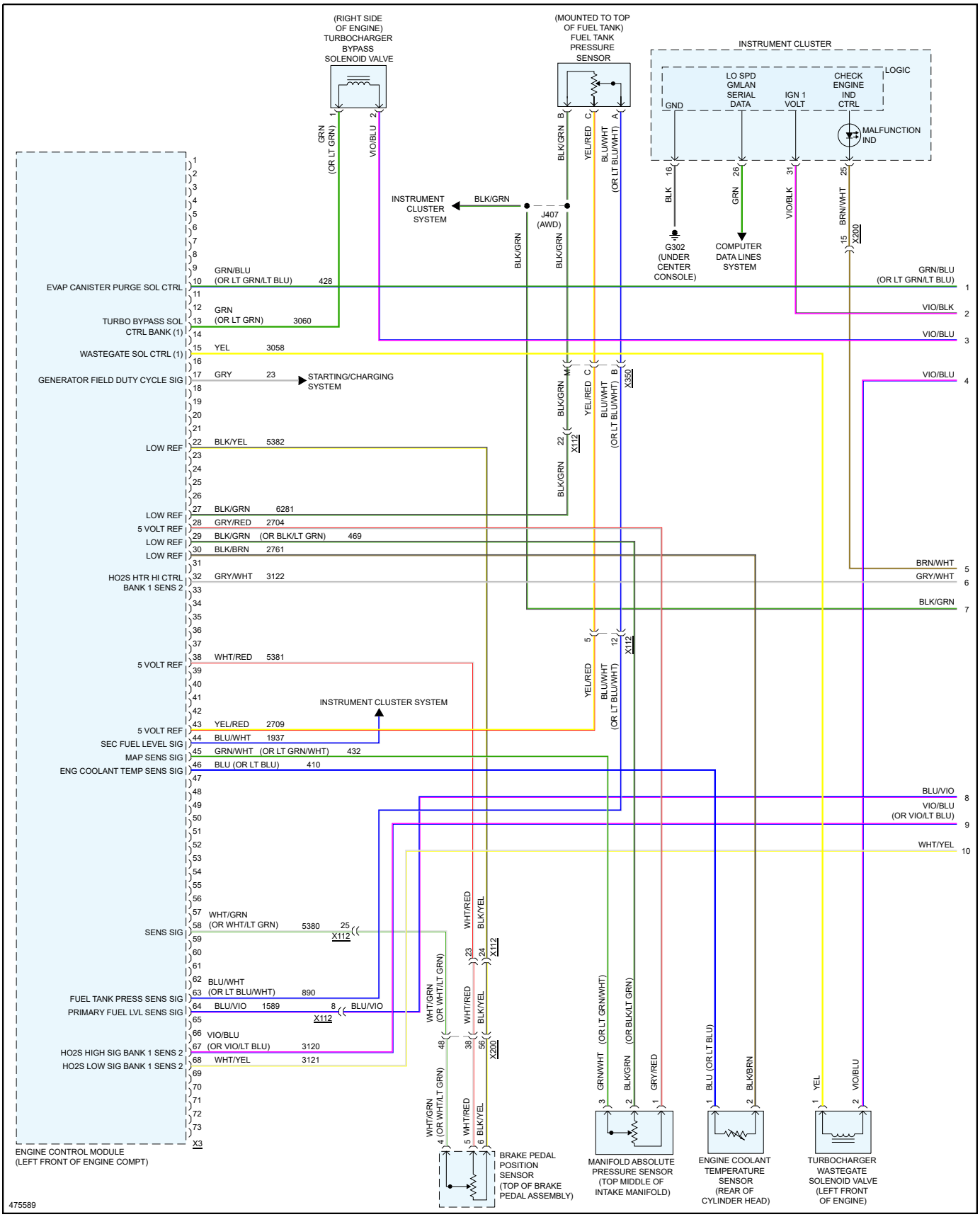
2015 Buick Truck Encore AWD L4-1.4L Turbo

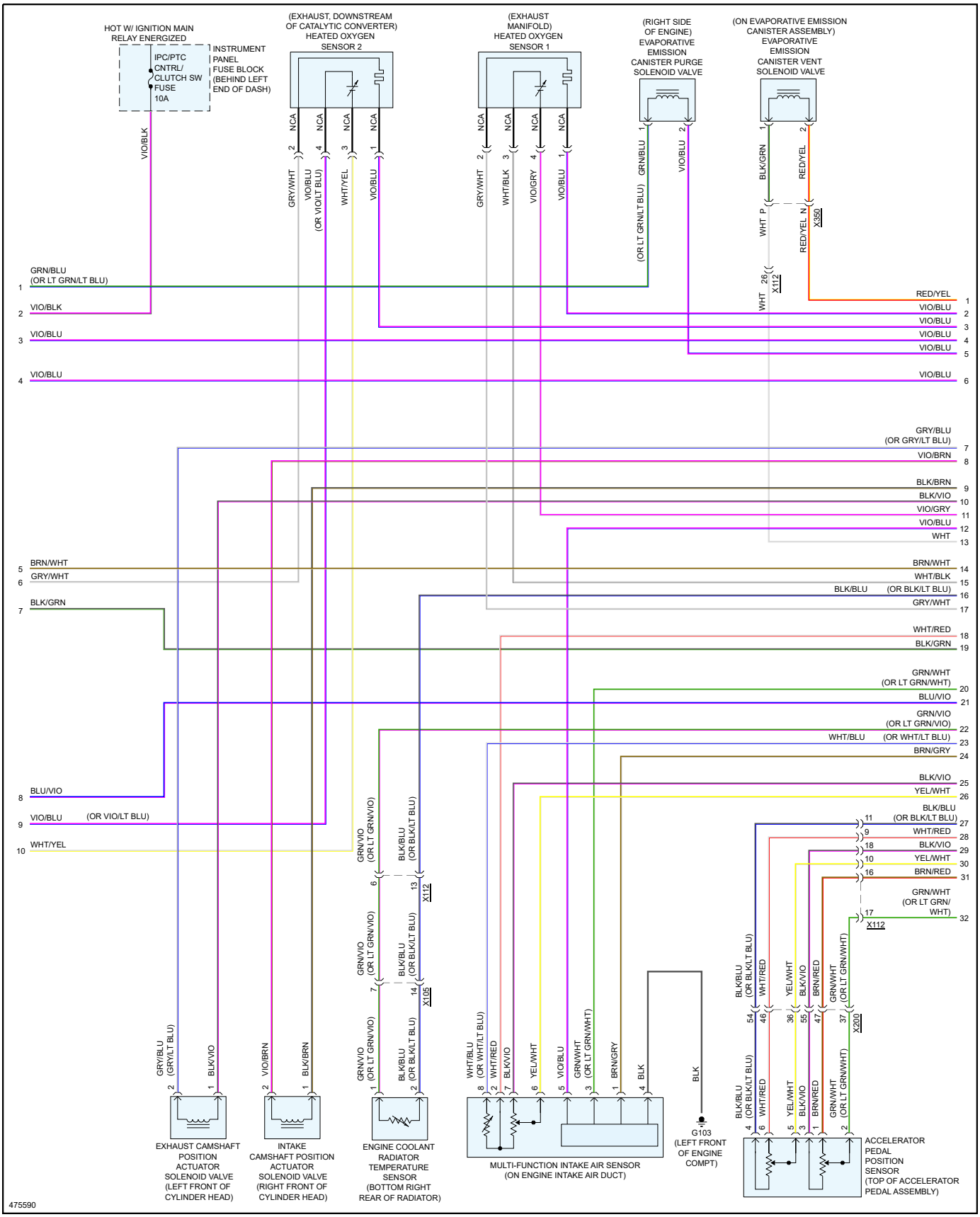
Vehicle > Powertrain Management > Diagrams > Electrical - Interactive Color (Non OE)

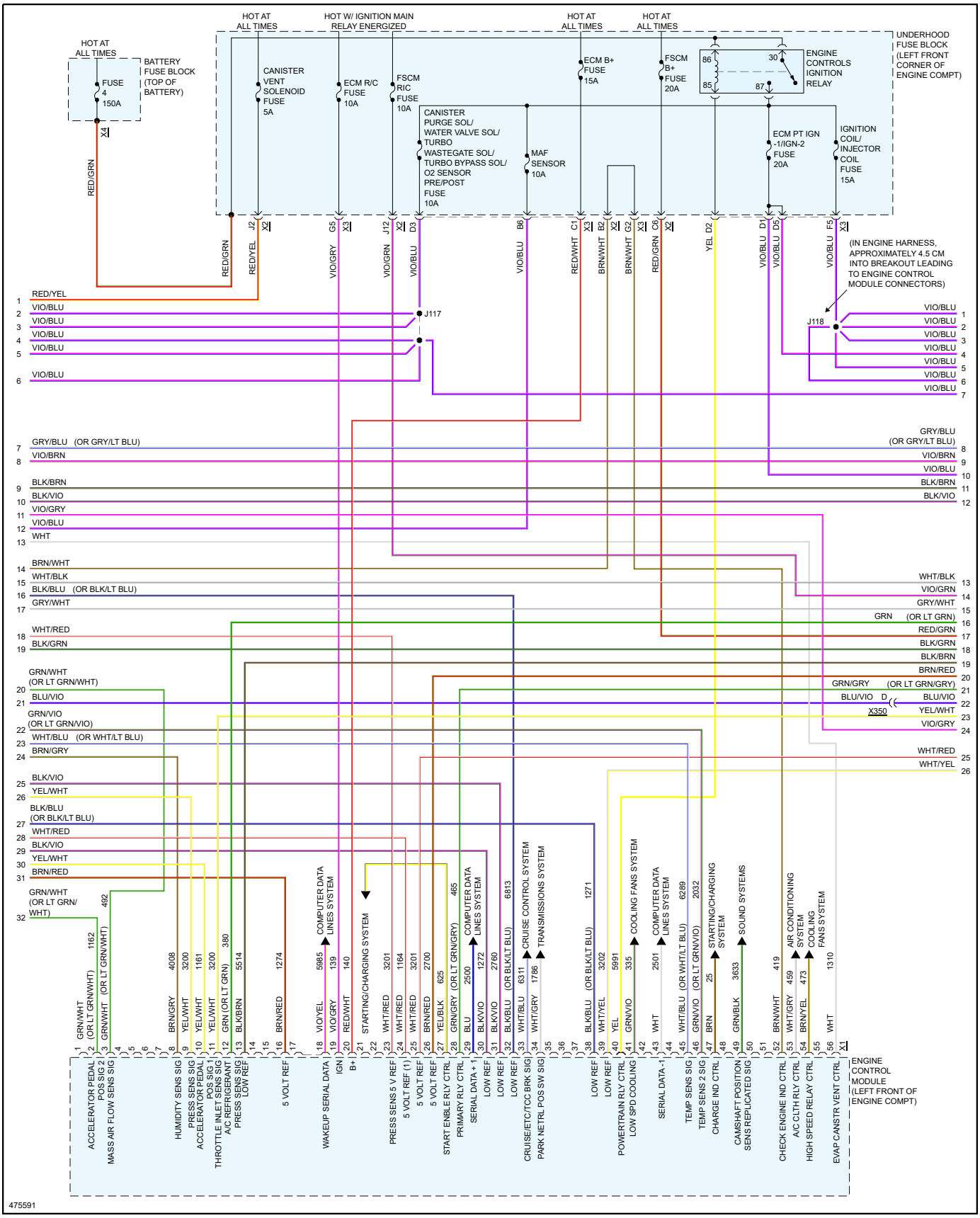
ENGINE CONTROLS

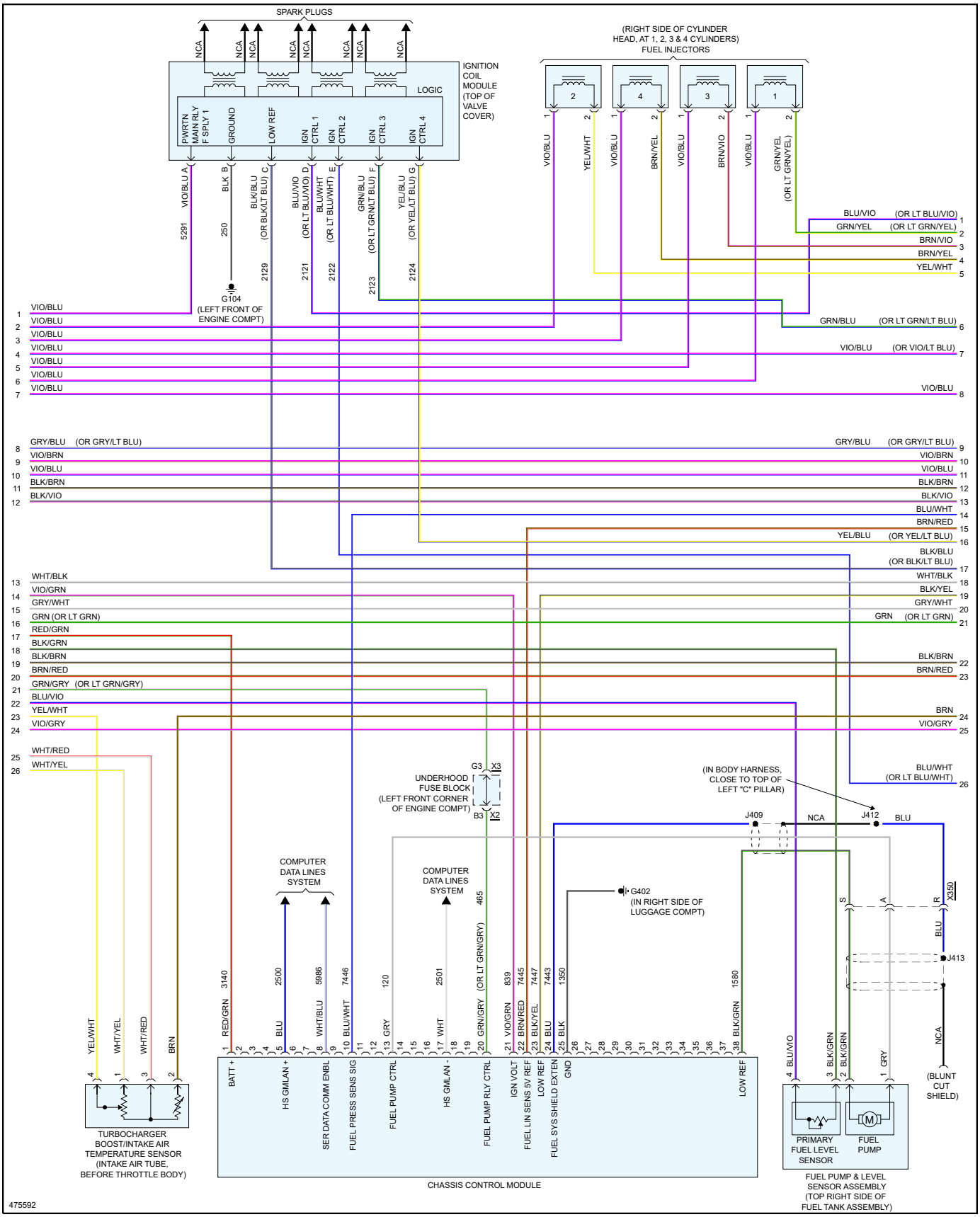
Engine Controls

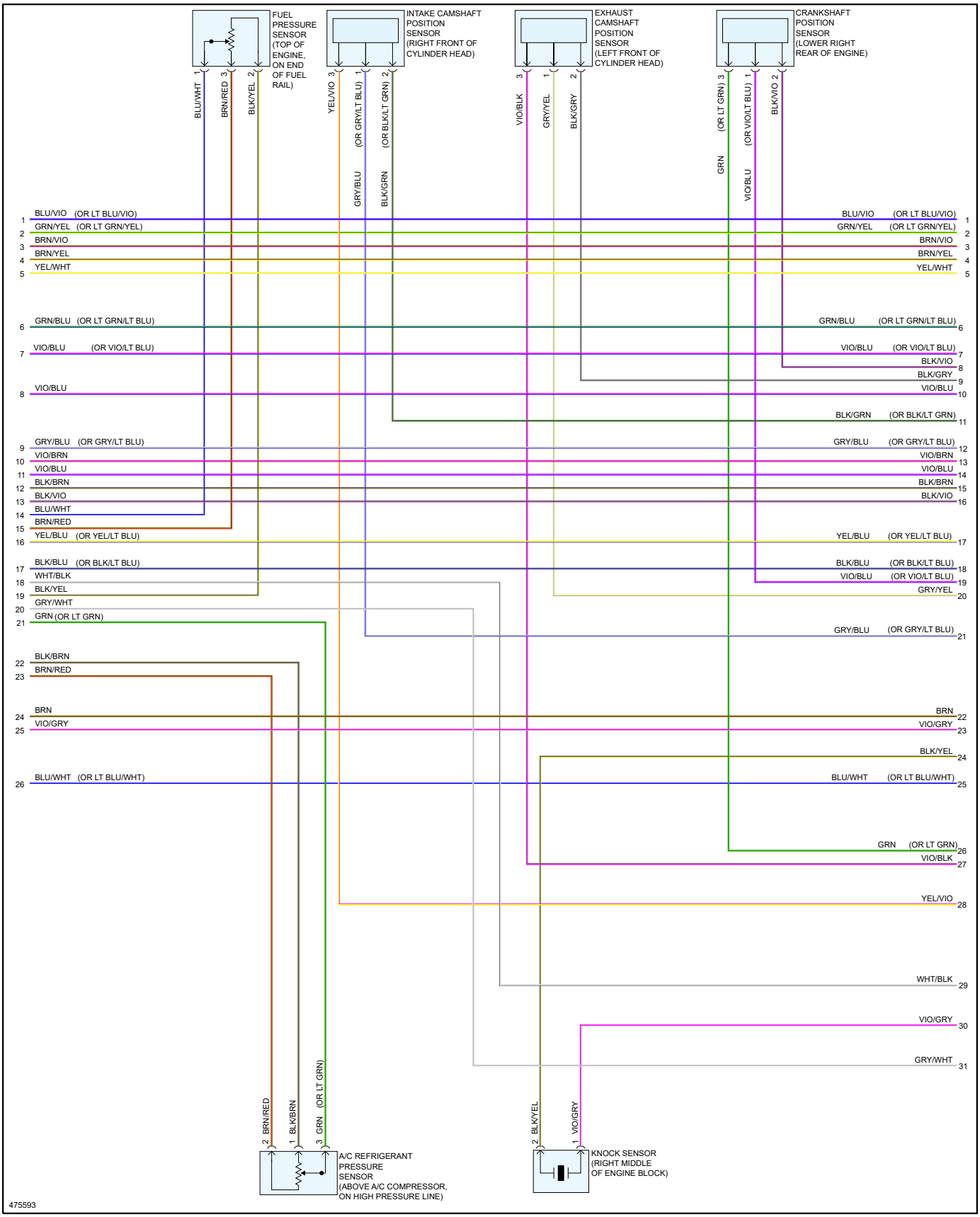
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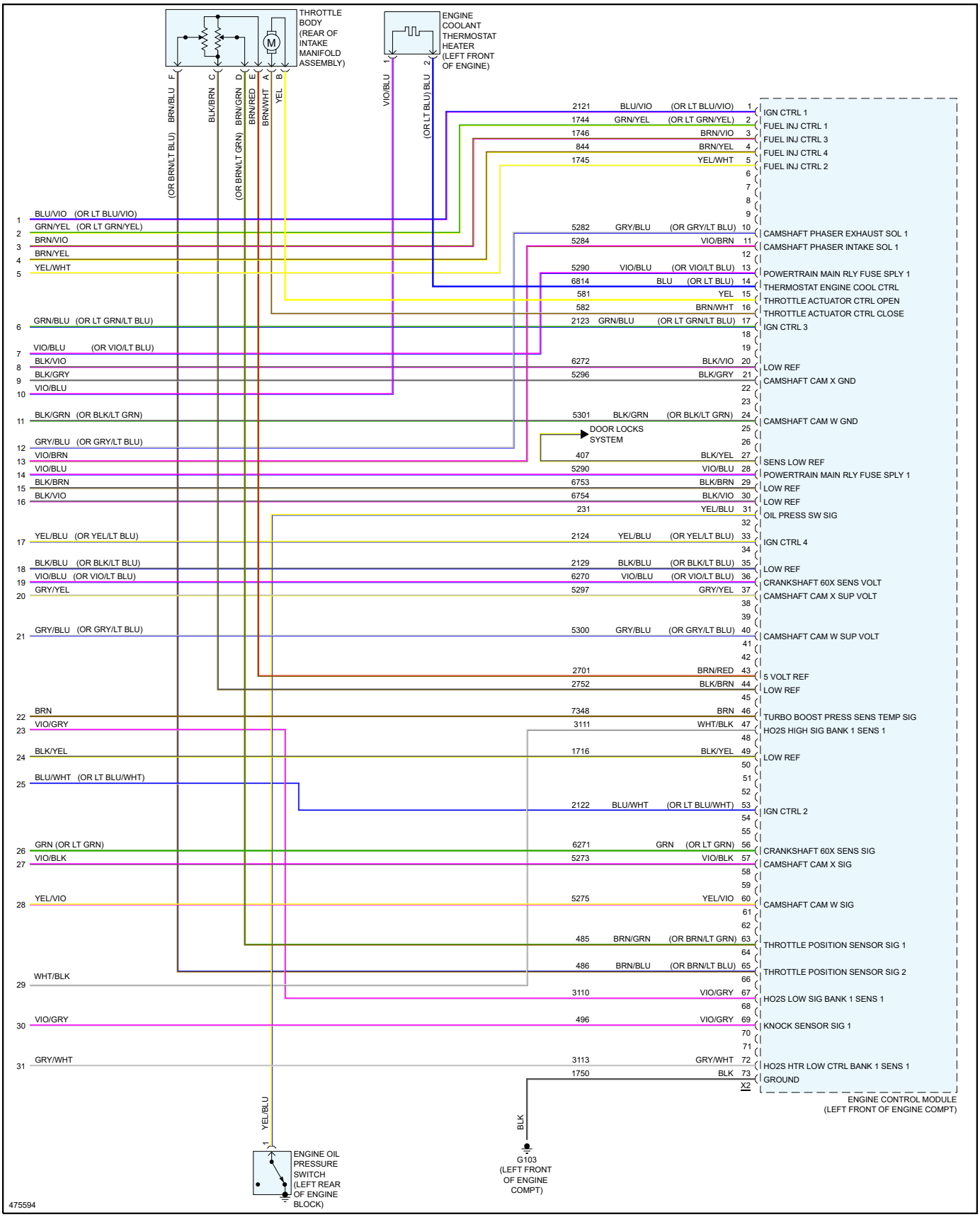

(RIGHT SIDE OF ENGINE) TURBOCHARGER BYPASS SOLENOID VALVE
(MOUNTED TO TOP OF FUEL TANK) FUEL TANK PRESSURE SENSOR
INSTRUMENT CLUSTER
LOGIC
LO SPD GMLAN SERIAL DATA
IGN 1 VOLT
CHECK ENGINE IND CTRL
GND
MALFUNCTION IND
INSTRUMENT CLUSTER SYSTEM
J407 (AWD)
G302 (UNDER CENTER CONSOLE)
COMPUTER DATA LINES SYSTEM
EVAP CANISTER PURGE SOL CTRL
TURBO BYPASS SOL CTRL BANK (1)
WASTEGATE SOL CTRL (1)
GENERATOR FIELD DUTY CYCLE SIG
STARTING/CHARGING SYSTEM
LOW REF
5 VOLT REF
SEC FUEL LEVEL SIG
MAP SENS SIG
ENG COOLANT TEMP SENS SIG
SENS SIG
FUEL TANK PRESS SENS SIG
PRIMARY FUEL LVL SENS SIG
HO2S HTR HI CTRL BANK 1 SENS 2
HO2S HIGH SIG BANK 1 SENS 2
HO2S LOW SIG BANK 1 SENS 2
ENGINE CONTROL MODULE (LEFT FRONT OF ENGINE COMPT)
BRAKE PEDAL POSITION SENSOR (TOP OF BRAKE PEDAL ASSEMBLY)
MANIFOLD ABSOLUTE PRESSURE SENSOR (TOP MIDDLE OF INTAKE MANIFOLD)
ENGINE COOLANT TEMPERATURE SENSOR (REAR OF CYLINDER HEAD)
TURBOCHARGER WASTEGATE SOLENOID VALVE (LEFT FRONT OF ENGINE)
475589


HOT W/ IGNITION MAIN RELAY ENERGIZED
IPC/PTC CNTRL/ CLUTCH SW FUSE 10A
INSTRUMENT PANEL FUSE BLOCK (BEHIND LEFT END OF DASH)
(EXHAUST, DOWNSTREAM OF CATALYTIC CONVERTER) HEATED OXYGEN SENSOR 2
(EXHAUST MANIFOLD) HEATED OXYGEN SENSOR 1
(RIGHT SIDE OF ENGINE) EVAPORATIVE EMISSION CANISTER PURGE SOLENOID VALVE
(ON EVAPORATIVE EMISSION CANISTER ASSEMBLY) EVAPORATIVE EMISSION CANISTER VENT SOLENOID VALVE
GRN/BLU (OR LT GRN/LT BLU)
VIO/BLK
VIO/BLU
VIO/BLU
VIO/BLU
RED/YEL
VIO/BLU
VIO/BLU
VIO/BLU
GRY/BLU (OR GRY/LT BLU)
VIO/BRN
BLK/BRN
BLK/VIO
VIO/GRY
VIO/BLU
WHT
BRN/WHT
GRY/WHT
BRN/WHT
WHT/BLK
BLK/BLU (OR BLK/LT BLU)
GRY/WHT
BLK/GRN
WHT/RED
BLK/GRN
GRN/WHT (OR LT GRN/WHT)
BLU/VIO
GRN/VIO (OR LT GRN/VIO)
WHT/BLU (OR WHT/LT BLU)
BRN/GRY
BLU/VIO
VIO/BLU (OR VIO/LT BLU)
WHT/YEL
BLK/VIO
YEL/WHT
BLK/BLU (OR BLK/LT BLU)
WHT/RED
BLK/VIO
YEL/WHT
BRN/RED
GRN/WHT (OR LT GRN/ WHT)
X112
X105
X200
X350
EXHAUST CAMSHAFT POSITION ACTUATOR SOLENOID VALVE (LEFT FRONT OF CYLINDER HEAD)
INTAKE CAMSHAFT POSITION ACTUATOR SOLENOID VALVE (RIGHT FRONT OF CYLINDER HEAD)
ENGINE COOLANT RADIATOR TEMPERATURE SENSOR (BOTTOM RIGHT REAR OF RADIATOR)
MULTI-FUNCTION INTAKE AIR SENSOR (ON ENGINE INTAKE AIR DUCT)
G103 (LEFT FRONT OF ENGINE COMPT)
ACCELERATOR PEDAL POSITION SENSOR (TOP OF ACCELERATOR PEDAL ASSEMBLY)
475590


HOT AT ALL TIMES
HOT W/ IGNITION MAIN RELAY ENERGIZED
HOT AT ALL TIMES
HOT AT ALL TIMES
BATTERY FUSE BLOCK (TOP OF BATTERY)
FUSE 4 150A
CANISTER VENT SOLENOID FUSE 5A
ECM R/C FUSE 10A
FSCM RIC FUSE 10A
CANISTER PURGE SOL/ WATER VALVE SOL/ TURBO WASTEGATE SOL/ TURBO BYPASS SOL/ O2 SENSOR PRE/POST FUSE 10A
MAF SENSOR 10A
ECM B+ FUSE 15A
FSCM B+ FUSE 20A
ENGINE CONTROLS IGNITION RELAY
ECM PT IGN -1/IGN-2 FUSE 20A
IGNITION COIL/ INJECTOR COIL FUSE 15A
UNDERHOOD FUSE BLOCK (LEFT FRONT CORNER OF ENGINE COMPT)
(IN ENGINE HARNESS, APPROXIMATELY 4.5 CM INTO BREAKOUT LEADING TO ENGINE CONTROL MODULE CONNECTORS)
J117
J118
X350
ENGINE CONTROL MODULE (LEFT FRONT OF ENGINE COMPT)
ACCELERATOR PEDAL POS SIG 2
MASS AIR FLOW SENS SIG
HUMIDITY SENS SIG
PRESS SENS SIG
ACCELERATOR PEDAL POS SIG 1
THROTTLE INLET SENS SIG
A/C REFRIGERANT PRESS SENS SIG
LOW REF
5 VOLT REF
WAKEUP SERIAL DATA
IGN
B+
STARTING/CHARGING SYSTEM
PRESS SENS 5 V REF
5 VOLT REF (1)
5 VOLT REF
START ENBLE RLY CTRL
PRIMARY RLY CTRL
SERIAL DATA + 1
CRUISE/ETC/TCC BRK SIG
PARK NETRL POS SW SIG
POWERTRAIN RLY CTRL
LOW SPD COOLING
SERIAL DATA -1
TEMP SENS SIG
TEMP SENS 2 SIG
CHARGE IND CTRL
CAMSHAFT POSITION SENS REPLICATED SIG
CHECK ENGINE IND CTRL
A/C CLTH RLY CTRL
HIGH SPEED RELAY CTRL
EVAP CANSTR VENT CTRL
COMPUTER DATA LINES SYSTEM
STARTING/CHARGING SYSTEM
CRUISE CONTROL SYSTEM
TRANSMISSIONS SYSTEM
COOLING FANS SYSTEM
SOUND SYSTEMS
AIR CONDITIONING SYSTEM
475591


SPARK PLUGS
IGNITION COIL MODULE (TOP OF VALVE COVER)
LOGIC
PWRTN MAIN RLY F SPLY 1
GROUND
LOW REF
IGN CTRL 1
IGN CTRL 2
IGN CTRL 3
IGN CTRL 4
(RIGHT SIDE OF CYLINDER HEAD, AT 1, 2, 3 & 4 CYLINDERS) FUEL INJECTORS
G104 (LEFT FRONT OF ENGINE COMPT)
G3 UNDERHOOD FUSE BLOCK (LEFT FRONT CORNER OF ENGINE COMPT)
(IN BODY HARNESS, CLOSE TO TOP OF LEFT "C" PILLAR)
J409
J412
G402 (IN RIGHT SIDE OF LUGGAGE COMPT)
J413
COMPUTER DATA LINES SYSTEM
TURBOCHARGER BOOST/INTAKE AIR TEMPERATURE SENSOR (INTAKE AIR TUBE, BEFORE THROTTLE BODY)
CHASSIS CONTROL MODULE
PRIMARY FUEL LEVEL SENSOR
FUEL PUMP
FUEL PUMP & LEVEL SENSOR ASSEMBLY (TOP RIGHT SIDE OF FUEL TANK ASSEMBLY)
(BLUNT CUT SHIELD)
475592


FUEL PRESSURE SENSOR (TOP OF ENGINE, ON END OF FUEL RAIL)
INTAKE CAMSHAFT POSITION SENSOR (RIGHT FRONT OF CYLINDER HEAD)
EXHAUST CAMSHAFT POSITION SENSOR (LEFT FRONT OF CYLINDER HEAD)
CRANKSHAFT POSITION SENSOR (LOWER RIGHT REAR OF ENGINE)
BLU/VIO (OR LT BLU/VIO)
GRN/YEL (OR LT GRN/YEL)
BRN/VIO
BRN/YEL
YEL/WHT
GRN/BLU (OR LT GRN/LT BLU)
VIO/BLU (OR VIO/LT BLU)
BLK/VIO
BLK/GRY
BLK/GRN (OR BLK/LT GRN)
GRY/BLU (OR GRY/LT BLU)
VIO/BRN
BLK/BRN
YEL/BLU (OR YEL/LT BLU)
BLK/BLU (OR BLK/LT BLU)
WHT/BLK
GRY/YEL
GRY/WHT
GRN (OR LT GRN)
BRN
BRN/RED
VIO/GRY
BLU/WHT (OR LT BLU/WHT)
YEL/VIO
A/C REFRIGERANT PRESSURE SENSOR (ABOVE A/C COMPRESSOR, ON HIGH PRESSURE LINE)
KNOCK SENSOR (RIGHT MIDDLE OF ENGINE BLOCK)
475593


THROTTLE BODY (REAR OF INTAKE MANIFOLD ASSEMBLY)
ENGINE COOLANT THERMOSTAT HEATER (LEFT FRONT OF ENGINE)
F BRN/BLU (OR BRN/LT BLU)
C BLK/BRN
D BRN/GRN (OR BRN/LT GRN)
E BRN/RED
A BRN/WHT
B YEL
1 VIO/BLU
2 (OR LT BLU) BLU
1 BLU/VIO (OR LT BLU/VIO)
2 GRN/YEL (OR LT GRN/YEL)
3 BRN/VIO
4 BRN/YEL
5 YEL/WHT
6 GRN/BLU (OR LT GRN/LT BLU)
7 VIO/BLU (OR VIO/LT BLU)
8 BLK/VIO
9 BLK/GRY
10 VIO/BLU
11 BLK/GRN (OR BLK/LT GRN)
12 GRY/BLU (OR GRY/LT BLU)
13 VIO/BRN
14 VIO/BLU
15 BLK/BRN
16 BLK/VIO
17 YEL/BLU (OR YEL/LT BLU)
18 BLK/BLU (OR BLK/LT BLU)
19 VIO/BLU (OR VIO/LT BLU)
20 GRY/YEL
21 GRY/BLU (OR GRY/LT BLU)
22 BRN
23 VIO/GRY
24 BLK/YEL
25 BLU/WHT (OR LT BLU/WHT)
26 GRN (OR LT GRN)
27 VIO/BLK
28 YEL/VIO
29 WHT/BLK
30 VIO/GRY
31 GRY/WHT
2121 BLU/VIO (OR LT BLU/VIO) 1 IGN CTRL 1
1744 GRN/YEL (OR LT GRN/YEL) 2 FUEL INJ CTRL 1
1746 BRN/VIO 3 FUEL INJ CTRL 3
844 BRN/YEL 4 FUEL INJ CTRL 4
1745 YEL/WHT 5 FUEL INJ CTRL 2
5282 GRY/BLU (OR GRY/LT BLU) 10 CAMSHAFT PHASER EXHAUST SOL 1
5284 VIO/BRN 11 CAMSHAFT PHASER INTAKE SOL 1
5290 VIO/BLU (OR VIO/LT BLU) 13 POWERTRAIN MAIN RLY FUSE SPLY 1
6814 BLU (OR LT BLU) 14 THERMOSTAT ENGINE COOL CTRL
581 YEL 15 THROTTLE ACTUATOR CTRL OPEN
582 BRN/WHT 16 THROTTLE ACTUATOR CTRL CLOSE
2123 GRN/BLU (OR LT GRN/LT BLU) 17 IGN CTRL 3
6272 BLK/VIO 20 LOW REF
5296 BLK/GRY 21 CAMSHAFT CAM X GND
5301 BLK/GRN (OR BLK/LT GRN) 24 CAMSHAFT CAM W GND
DOOR LOCKS SYSTEM
407 BLK/YEL 27 SENS LOW REF
5290 VIO/BLU 28 POWERTRAIN MAIN RLY FUSE SPLY 1
6753 BLK/BRN 29 LOW REF
6754 BLK/VIO 30 LOW REF
231 YEL/BLU 31 OIL PRESS SW SIG
2124 YEL/BLU (OR YEL/LT BLU) 33 IGN CTRL 4
2129 BLK/BLU (OR BLK/LT BLU) 35 LOW REF
6270 VIO/BLU (OR VIO/LT BLU) 36 CRANKSHAFT 60X SENS VOLT
5297 GRY/YEL 37 CAMSHAFT CAM X SUP VOLT
5300 GRY/BLU (OR GRY/LT BLU) 40 CAMSHAFT CAM W SUP VOLT
2701 BRN/RED 43 5 VOLT REF
2752 BLK/BRN 44 LOW REF
7348 BRN 46 TURBO BOOST PRESS SENS TEMP SIG
3111 WHT/BLK 47 HO2S HIGH SIG BANK 1 SENS 1
1716 BLK/YEL 49 LOW REF
2122 BLU/WHT (OR LT BLU/WHT) 53 IGN CTRL 2
6271 GRN (OR LT GRN) 56 CRANKSHAFT 60X SENS SIG
5273 VIO/BLK 57 CAMSHAFT CAM X SIG
5275 YEL/VIO 60 CAMSHAFT CAM W SIG
485 BRN/GRN (OR BRN/LT GRN) 63 THROTTLE POSITION SENSOR SIG 1
486 BRN/BLU (OR BRN/LT BLU) 65 THROTTLE POSITION SENSOR SIG 2
3110 VIO/GRY 67 HO2S LOW SIG BANK 1 SENS 1
496 VIO/GRY 69 KNOCK SENSOR SIG 1
3113 GRY/WHT 72 HO2S HTR LOW CTRL BANK 1 SENS 1
1750 BLK 73 GROUND
X2
ENGINE CONTROL MODULE (LEFT FRONT OF ENGINE COMPT)
YEL/BLU
1
ENGINE OIL PRESSURE SWITCH (LEFT REAR OF ENGINE BLOCK)
BLK
G103 (LEFT FRONT OF ENGINE COMPT)
475594